

done at the same time without separate equipment, according to *sensefly.com*.

Chasco Constructors is another company that started using drone technology for construction and site monitoring. For one of the projects, they used DJI Phantom 3 Pro drone and drone-mapping software from DroneDeploy, which resulted in the completion of the project before the estimated schedule. According to data sourced from *dronedeploy.com*, the total budget for the construction of the 6,500 square-

metre structure was 29 million dollars. The replacement of most manual tasks with drone mapping slashed the survey costs by 75 per cent.

Startups like DroneBase and Garuda Robotics also offer services like aerial surveying to architecture, engineering and construction (AEC) industry, among others.

Extreme temperatures and wind can affect drone operations. Another issue is that battery power may not be enough for complete inspection of big

sites. Also, since a malfunction such as a damaged propeller can cause the drone to crash, an operator should avoid flying over areas where people are present. A qualified person must pilot a commercial drone to prevent damage and necessary permissions need to be obtained. With enough precautions, drones are capable of shifting the overall paradigm in the construction sector and bring in innumerable benefits for the industry.

—Ayushee Sharma

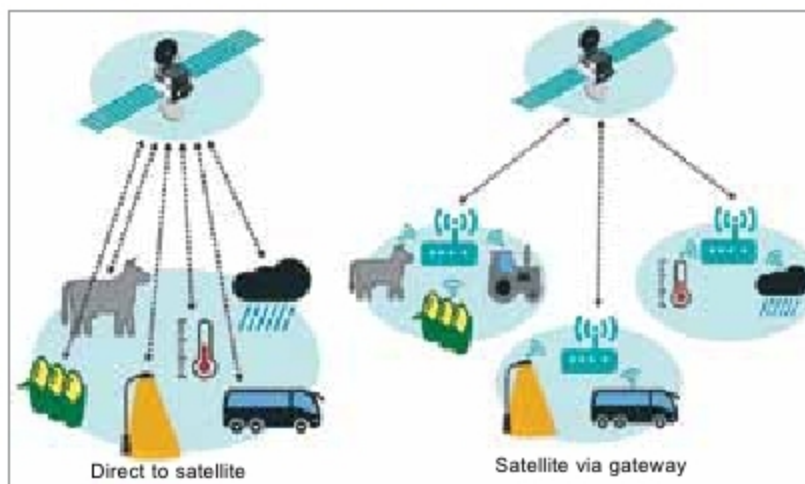
## 3 LATEST TECHNOLOGY IN SPACE RACE: SATELLITE IoT

**The Internet of Things (IoT) is now globally proliferating to intelligently connect devices in order to meet increasing consumer demands as well as improve productivity. IoT further helps to increase the accuracy of data-based forecasts and trends**

From home to industrial applications, the deployment of Internet of things (IoT) systems is globally proliferating to intelligently connect devices and gaining useful information in an attempt to meet increasing consumer demands as well as improve productivity. As the total number of connected sensors increases, so do the data points, thereby increasing the accuracy of data-based forecasts and trends.

This requires consistent and ubiquitous coverage to reach even the remotest assets so that every business gets a reliable IoT connectivity service. For this, satellites are a suitable option to connect sensors and IoT devices to the cloud and communicate even in harsh weather conditions (via L-band).

Unlike most of the present terrestrial infrastructure options that are complex, satellite connectivity enhances coverage in areas with limited or no cellular and Wi-Fi connectivity along with easier device management via an IoT connectivity platform. For example, Iridium Edge is a satellite IoT communications device that can be rapidly deployed and complements terrestrial based solutions. The company's satellite cloud-based



(Credit: [www.ttp.com](http://www.ttp.com))

solution CloudConnect offers global IoT coverage through Amazon Web Services (AWS) and is a singular communications platform to manage connected devices.

In direct configuration mode, satellite connectivity can either be used only when necessary in conjunction with terrestrial IoT access networks (like cellular, LPWAN), or be solely used where it is the only possible means of connectivity for transmitting data. Hybrid terrestrial satellite systems result in lower cost due to terrestrial-only IoT devices, while satellite-connected terminals ensure coverage. Myriota is one such startup that provides direct satellite connectivity for IoT uses, for purposes like environment monitoring. Its advantages include security and scalability.

For sensors and devices that do not

use much data, a direct satellite connection can be expensive in terms of equipment and power consumption. In backhaul configuration, satellite connects to a tower which then connects to devices via other suitable wireless communication technologies.

Since the deployment needs to be done in areas that are off-grid or have an unreliable power supply, the service needs to be designed for low power

operation so that renewable energies like solar can suffice. The deployment should be preferably flexible enough to match hybrid connectivity needs. Apart from this, maintenance can be made easy with over-the-air (OTA) firmware upgrades.

Low Earth orbit (LEO) maintains low power consumption and is a cost-effective option. It is also well-suited to narrowband requirements of IoT but needs steerable antennas due to moving satellites. On the other hand, geostationary orbit (GEO) satellites need large terminal antennas and sufficient directivity to prevent interference with satellites in their vicinity.

This opens a plethora of opportunities for satellite operators like deploying new LEO or HEO (highly elliptical orbit) constellations and selling additional capacity in different bands (like